

Grading and evaluation

The grading and evaluation of gems can start even before they are removed from the ground. Within some gem deposits, certain areas are known to yield more or better-quality gem material, and are thus mined first. Only a small percentage of what is recovered is actually of gem quality, and this is further sorted to separate out the gem material. Any gem-quality roughs found are then carefully evaluated for colour, clarity,

and size. Even much of this selected material may remain uncut if it is too small, oddly shaped, or for some reason does not fit the current market demand. Cutting is expensive and time-consuming, and so meticulous grading at this early stage is essential. To be certain, the cutter will make his or her final decision about what is cut, but the greater the evaluation and grading before gems reach the cutter's workshop, the better.

Non-gem quality

The specimen shown here is of ordinary zoisite, and is typical of virtually all zoisite material. It is opaque, and even when well crystallized, it is not of gem quality. Until the discovery of tanzanite, zoisite's gem-quality variant (see p.253), only a tiny amount of gem-quality zoisite was known.



Non-gemmy material

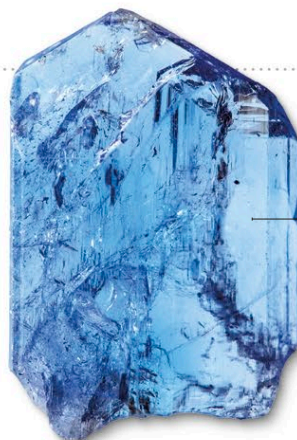
Zoisite crystal This zoisite rough is not of gem quality and would be of value purely as a mineral sample.

**The jewel...
is concentrated
brilliancy, the
quintessence
of light**

Charles **Blanc**
19th-century author

Mid-gem quality

Because tanzanite is a relatively uncommon stone, even medium-quality gems are valuable enough to facet. In more common stones, such as amethyst, a mid-quality rough is seldom faceted because the cost of cutting it would exceed the final value of the finished gem.



Internal flaws and inclusions

Pale tanzanite rough This rough crystal is gem quality, but has a number of small flaws and inclusions.

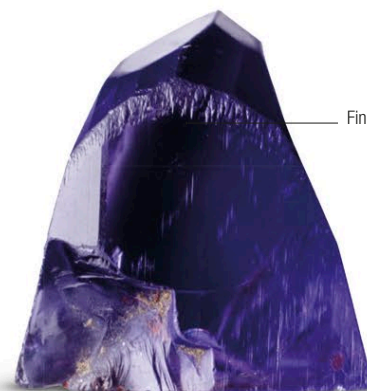


Facets conceal imperfect interior

Cut pale tanzanite Faceting can be used to conceal or disguise internal imperfections in mid-quality gems, as seen in this example.

Superior quality

A relatively small portion of even a select gemstone rough is of superior quality. In this tanzanite example, both the rough and the cut stone are of visibly higher quality than the mid-quality grade (above).



Fine colouring

Superior-quality tanzanite rough Even as a rough, the quality of this tanzanite can be seen in its deep colour and clear interior.



Skilled faceting

Cut tanzanite The best-quality gemstones combine superior-grade material with highly skilled craftsmanship, as in this stone.